

AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

FLIGHT SIMULATION AND TESTING OF A NOVEL SYNCHROPTER CONCEPT FOR THE AERIAL-RECOVERY OF FALLING ROCKET STAGES



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Abstract

Implementing Xplane flight simulation software to generate a flight model, designed by the Wholly Reusable Launch System Recovery Team, to conduct stability and quantitative handling tests. Investigations into longitudinal, lateral and directional oscillatory behaviour upon a unit step control input found that the aircraft model, both with and without the slung load, was stable about these axes. However, it was found large sudden increases in collective resulted in a highly unstable mode. Whilst the model was easy to fly and generally very manoeuvrable, yaw control was an issue. The lack of detail that could be applied to the model in the software, as well as pilot inexperience was an issue that lead to inaccuracies in the results.

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Part I

Pre-Conceptual

1 Introduction

SpaceX recently conducted the first private launch of American astronauts to the International Space Station. Jeff Bezos's Blue Origin and Richard Brandon's Virgin Galactic both hope to start commercialised flights for Space Tourism. These companies all understand that the main problem to overcome in this industry is cost and hence re-usable rockets are the key to being able to be competitive in this lucrative sector [2]. Third year aeronautics students at Imperial College London have consequently undertaken an investigation to design an orbital launch system that can be completely recovered following a launch and re-used [3]. The Wholly Reusable Recovery Team was tasked with the objective of designing an aerial vehicle that can capture the falling rocket stages in mid-air, retrieving them safely.

Being assigned to the Flight Simulation and Testing Sub-team, we were tasked with generating a model of the designed aerial vehicle in an appropriate simulation software and subsequently testing the manoeuvrability, stability and capture procedure to ensure that the given design was able to operate as laid out in the design brief. Using flight simulating software to computationally test how an aircraft will behave in real life is a very important step in the design process. It allows engineers to validate their design cost effectively, being able to identify any instabilities or failures before they proceed to building a conceptual design.

2 Conceptual Design

The aerial vehicle was thought to be one of either three choices; helicopter, VTOL aircraft or a conventional plane. After deliberation, a helicopter was chosen as the most suitable choice, due to its higher maneuverability compared to an aircraft and its simplicity compared to a VTOL aircraft. Furthermore, it could conduct ship based operations, extending its range of operation as it could be deployed anywhere on the high seas. What it lacks in fuel efficiency, range and speed compared to VTOL and planes is outweighed by the large margin for error given it can conduct multiple attempts of recapture [4].

The baseline configuration consisted of [5]:

Table 1: Baseline Line Configuration Reasoning

Decision	Reason
2 Bladed Synchropter	Reduction in mechanical complexity Low disc loading resulting in large lifting capability and high ceiling
Unmanned	Reduction in weight due to lack of pilot and cabin Reduction in flight risk
Skids	Reduction in weight due to the absence of retracting mechanism

3 Software Choice

FlightGear and Xplane were the two obvious choices for flight simulation software. Since none of the team had built a flight model before and had a limited time frame, it was essential that the complexity in generating a model was not too high yet accuracy must not be compromised.

3.1 FlightGear

FlightGear is an open source simulation software that uses stability derivatives to generate a flight dynamic model from either JSBSim or YASim with a user input .xml file (coded in C++) which includes the necessary data about the design to be modelled. JSBSim will generate flight dynamics based on stability derivatives (which the Flight Mechanics team would be required to calculate) as well as wind tunnel data. However given that there is limited information on implementing it on a helicopter design this was ruled out. YASim generates a flight dynamics model from the geometry of the aircraft and a solver matches to known flight data [6]. A benefit of the open source nature of this software is that it is very flexible and hence the capture process could be modelled accurately.

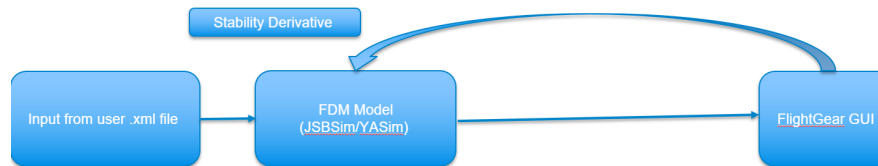


Figure 1: FlightGear Flight Dynamics Model [1]

3.2 Xplane

Xplane breaks down the geometry of the model using a finite elements method in 3 dimensions. Velocity vectors of each element are subsequently calculated. Propwash is found by observing the area and thrust of each propeller disk and using local air density uses momentum methods to find propwash required. 2D Airfoil data is extrapolated into 3D using finite wing methods. Compressible flow effects are corrected using Prandtl-Glauert. Finally dynamic pressures and forces are calculated allowing linear and angular acceleration to be calculated from mass and moment of inertia [7]. The user interface of Xplanemaker makes it easy to develop models in the software, however, this ease of use results in a lack of flexibility requiring external plugins to model the recapture process.

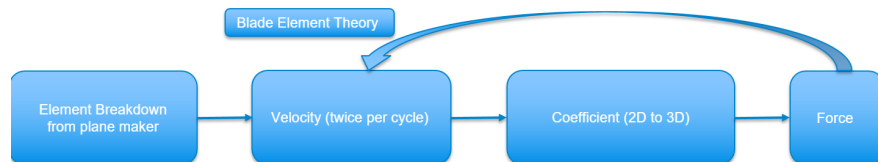


Figure 2: Xplane Flight Dynamics Model [1]

3.3 Decision

The simulation team chose Xplane for two main reasons. Firstly, blade element momentum theory is able to generate an accurate flight model without the requirement for wind tunnel data or complex C++ coding and requires less complex value and geometric inputs. Most importantly, despite

being less flexible, with Xplane we would be able to verify the flight data the flight mechanics team generate rather than relying on it to generate the flight model if FlightGear was chosen.

Part II

Simulation and Flight Testing

4 Simulation Building

The Xplane software comes with Xplanemaker, an easy to use user interface in which one can design the required aircraft from a set of value and user inputs. The simulation broke down the design into separate tasks. I was tasked with inputting the Fuselage, Weight and Balance and Landing Gear Parameters.

4.1 Fuselage

Unfortunately this task was not as simple as importing a CAD file of the fuselage. As shown in Figure 3, Xplanemaker requires a series of cross sections along the length of the fuselage, approximated by a series of points. This consequently generates panels used by the simulation in blade element theory calculations. Working closely with Tejasva Malhotra and Atri Sharma from the CAD team .dxf files were produced of 11 cross sections equidistant along the fuselage in which co-ordinates of the 9 points could be found. One simplification made was that the nose of the aircraft was rounded with a camera lens attached, however, in the interests of time and simplicity the nose was assumed to be a point, as I believed it would not have too much effect on the aerodynamics.

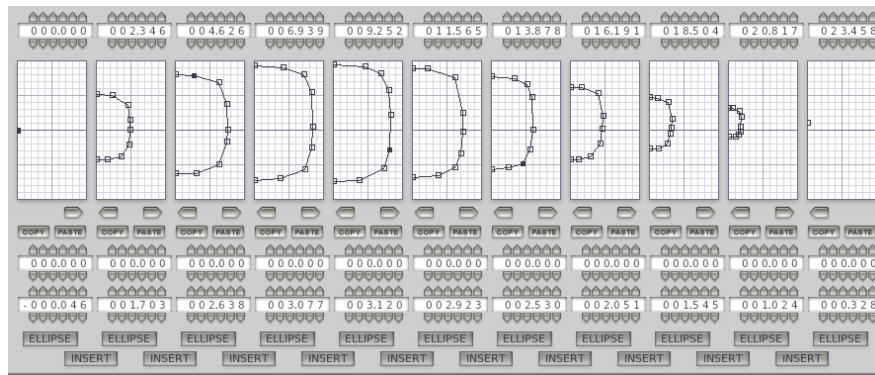


Figure 3: Xplanemaker User Interface for Fuselage Design

4.2 Weight and Balance

Setting the reference point at the tip of the nose the CG locations calculated by Atri Sharma could be inputted. As well as the size and location of the fuel tanks playing a key part in the change in weight and location of CG of the aircraft during flight. Interestingly, unlike my other simulation team members I had to have a fuel pressure in the tanks for there to be fuel flow into the main engine. A point I will later discuss in the Discussion.

4.3 Landing Gear

Since the landing skids were not finally completed before the simulation was built, I implemented skids that in terms of drag were identical. Hence, since we were not testing the landing of the helicopter this would not skew our results as our model was aerodynamically unaltered.



Figure 4: The Helicopter Model in the Simulation

5 Stability Testing

With the aircraft model completed, it could now be tested in Xplane to analyse its performance and stability and capture process. Undertaking the stability tests, I frequently referred to the Rotary Wing Manual, Naval Test Centre 1959. Despite being a slightly dated piece of literature, it clearly outlined multiple stability tests that could be conducted to investigate the aircraft's stability characteristics. These tests were conducted both with and without the slung payload to see if that had any drastic effect on the helicopter's stability and mainly perturbing it from equilibrium, this case from trimmed steady level flight. All tests were conducted between 4000-5000 ft asml since this is the region in which the aircraft would be operating for the majority of the time. Furthermore, it must be noted that since Xplane makes a very large amount of calculations per second there were arrays approximately 10 million large for a 40 second test phase, leading to some difficulties in data processing. Furthermore, Pitch is further defined as the longitudinal fuselage attitude and is not referring to the pitch angles of the rotor blades.

5.1 Static Longitudinal Stability

Positive static longitudinal stability is defined as when an aft longitudinal pull control force is required to hold the aircraft in a nose up attitude, along with a decreased airspeed and likewise if a push control force is required to hold the aircraft in a nose down attitude, along with increasing airspeed [8]. This was achieved by putting the aircraft in trimmed steady level forward flight, and slowly applying longitudinal control input forces recording the new trim speed. The collective remained at 50% throughout the test.

In Figure 5 longitudinal control input is defined as the user's elevator input, as a ratio to full upward deflection. In this instance, it has been normalised to show new control position relative

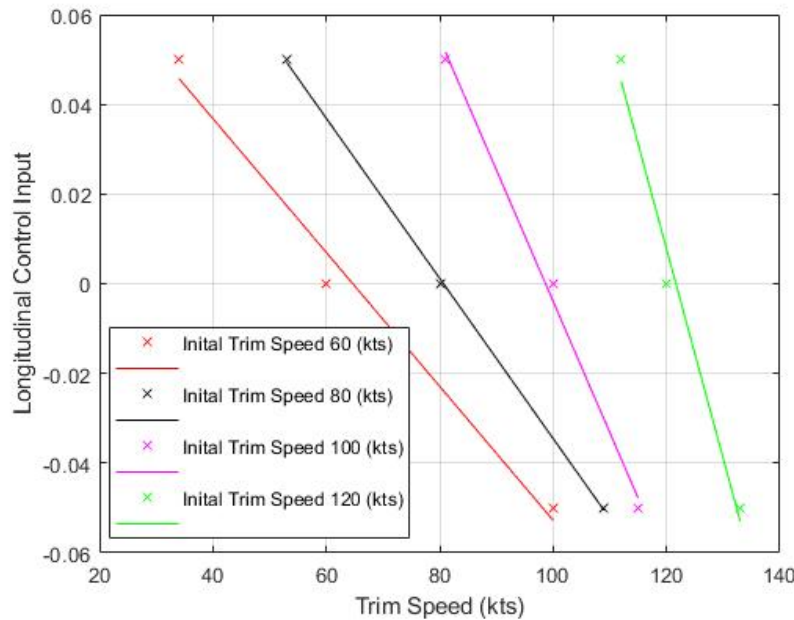


Figure 5: Static Stability

to new trim position. The different colours represent the 4 different initial trim speeds that were conducted. Figure 5 shows us that the aircraft is longitudinally statically stable as it satisfies the above condition. It is important to note the sensitivity of the control input since a small perturbation in the longitudinal control input resulted in a large change of airspeed. This was more dramatic at low speeds (one can see the increasing negative gradients of the linear regression lines) since aerodynamic drag is relatively low.

5.2 Dynamic Longitudinal Stability

This was an investigation into the aircraft's oscillation characteristics following a sudden longitudinal disturbance displacing it from trim. This was conducted by placing the aircraft in trimmed (100kts) steady level flight and applying a sudden longitudinal displacement. A time history of the ensuing oscillation is then recorded. The helicopter is defined as longitudinally stable if the oscillations decrease in amplitude over time. Oscillations about other axes are also investigated to see if there is any coupling about multiple axes, but no evidence of coupling were found. If you are unable to see the y labels, in descending order; Pitch (Degrees), Longitudinal Control Input, True Airspeed (kts) and Normal Acceleration. In Figure 6 a) we can see that with a small control input held for about a second produces damped longitudinal oscillations that indicated positive dynamics longitudinal stability. If we however increase the control input as shown in Figure 6 b) we can see the magnitude of the oscillations increase over time indicating to the contrary that the aircraft possesses negative stability. Looking at both graphs we can ignore the noise during the first 20 seconds since this was the pilot stabilising the aircraft after un-pausing the simulation.

Interestingly if you see appendix A.1 we can observe the effect on the performance of the engine to the longitudinal control input. In Figure 12 we see that whilst the engine response, in particular the RPM and Torque, slightly oscillate with the occasional flutter (indicating the r) they remain bounded, with the RPM varying by only ± 1 rad/s and Torque by about 500 Nm. If we look at the engine performance during the unstable oscillations (Figure 13), however, we see that although there

isn't much flutter the oscillations in RPM (± 5 rad/s) and Torque (± 2000 Nm) are unbounded and increase over time. One hypothesis is that this is caused by the low aspect ratio (short and fat) nature of the rotor blades leading them to be less efficient and experiencing higher drag forces with higher angles of incidence (which occur with a large displacement from equilibrium), which can put strain on the aircraft's engine causing the oscillatory effect. This could be further aggravated by how light the blades were, giving them low inertia and therefore transferring a lot more aerodynamic load onto the engine. Nevertheless, longer blades have lower ground clearance, which is already constrained due to synchropter's having angled rotors.

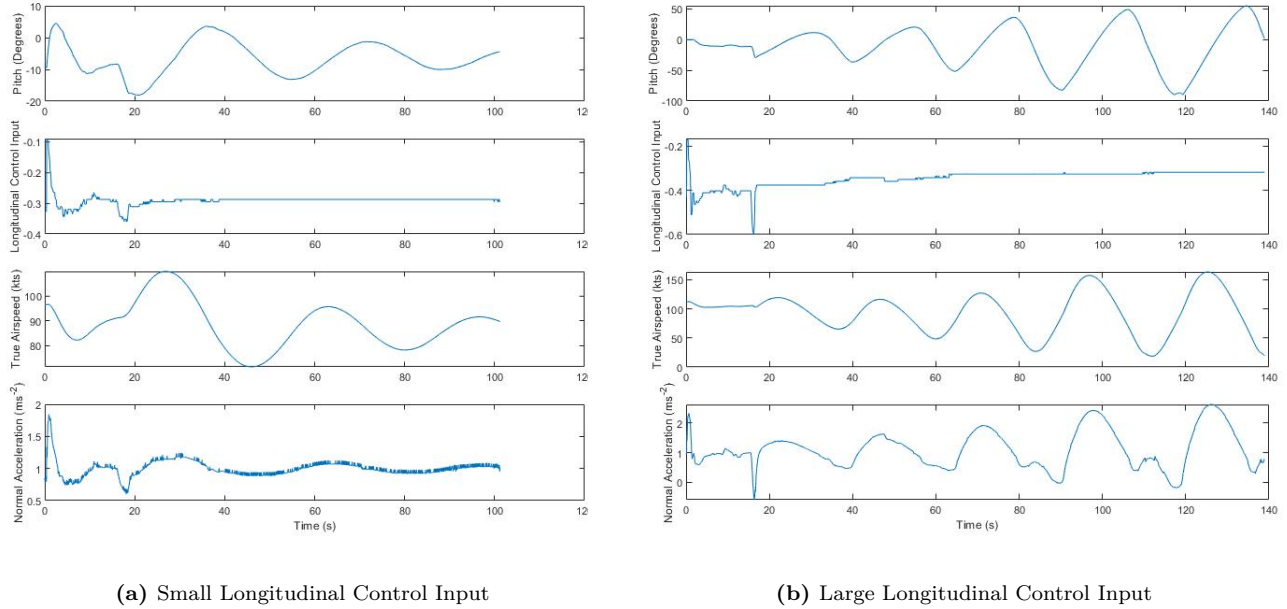


Figure 6: Longitudinal Oscillatory Response to Control Input

5.3 Dynamic Lateral Control Fixed Stability

The purpose of a dynamic lateral stability test is to find the ability to recover from lateral disturbances. This may occur with strong gusts or more appropriately the sudden lateral force experienced at the moment of capture. Similar to dynamic longitudinal stability, ensuing oscillations are observed and the aircraft is defined as stable if the lateral fuselage attitude converges. This test is conducted by providing a step lateral control impulse from trimmed steady level flight (100kts) and recording the following time history rolling oscillation. It must be noted at this point that, in the interest of time, the lateral static stability test was skipped, consequently we are assuming that the aircraft possesses stable gradient of lateral control position with respect to side slip [8].

One can see from Figure 7 that upon a step unit lateral control input the helicopter rolls to an angle of around 60 degrees before slowly returning back to its natural 0 degree rolling angle, acting as if it were a critically damped system. This indicates the aircraft is very laterally dynamically stable, which is a huge advantage since during the capture of the falling rocket stage it could experience large lateral forces. The time history of Yaw (degrees) is also plotted to investigate the coupling between yaw and roll, a phenomenon known as Dutch Roll. The discontinuity is due to the fact that when I conducted this test, Yaw is defined in relation to the ground rather than the aircraft so when I passed the origin I went from 360 to 0 degrees. Despite this, we can see that there is some

coupling effect indicating a degree of Dutch Roll. A poorly damped Dutch roll is to be avoided since the mode will be repeatedly excited by lateral gusts leading to instabilities [9], however, since the yaw and roll to be highly damped this mode is not a concern.

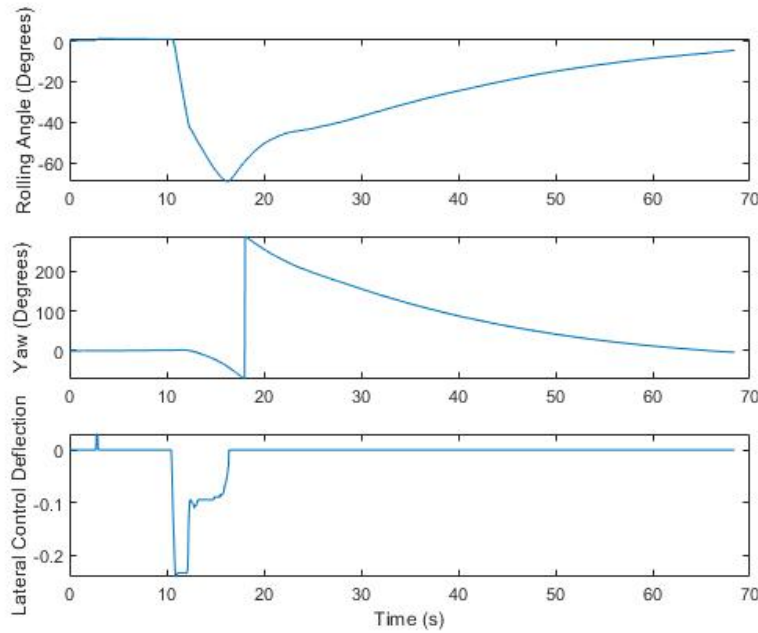


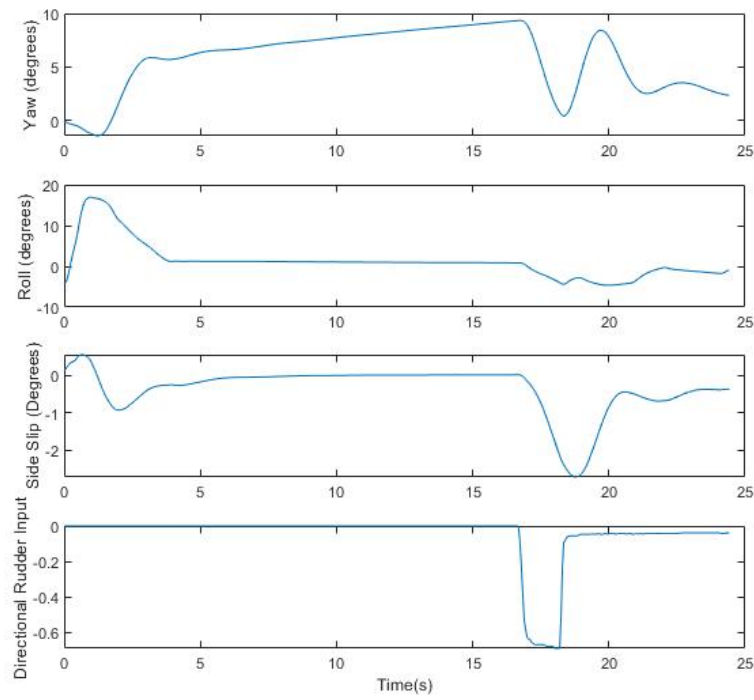
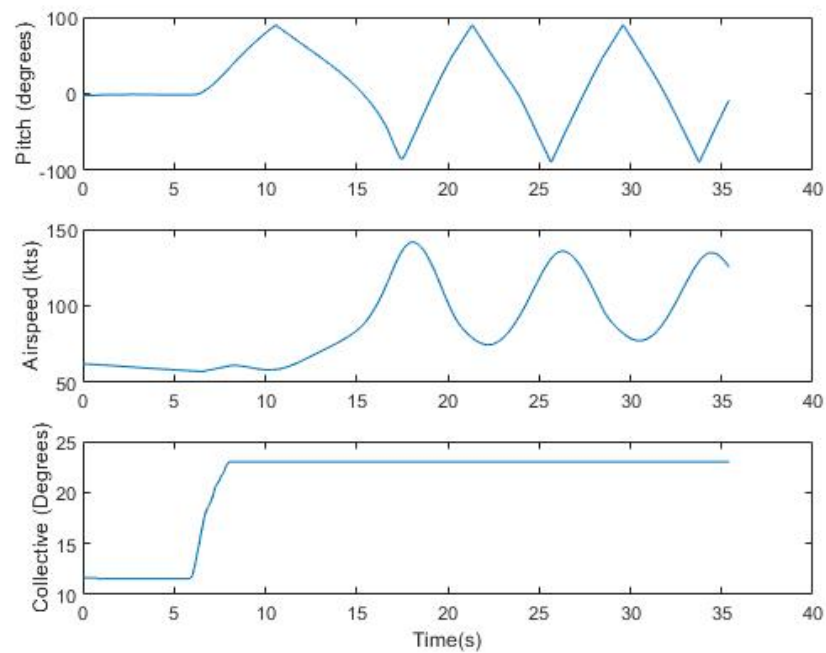
Figure 7: Lateral Step Impulse Response

5.4 Dynamic Directional Control Fixed Stability

Directional stability tests are investigations on effects of rudder inputs on heading direction. This test is conducted by putting the aircraft in steady level trimmed flight (100kts) and displacing the rudder pedals for a short period of time and, as before, obtaining time histories of yaw and side slip and if they converge the aircraft possesses positive dynamic directional stability [8]. From Figure 8 one can see that a large rudder input does result in oscillations, but their magnitude quickly dissipates, indicating directional stability. More interestingly however, if we look between 5 to 15 seconds we can see that with no rudder control input the aircraft was still slightly yawing at high speeds. Further investigation revealed no discrepancy between the two rotor RPMs which led me to believe perhaps the rotor wake was disturbing flow over the horizontal stabilisers. Investigating this using the flow field view in Xplane (see Appendix A.2) again no evidence was found that the rotor down wash was affecting the flow over the horizontal stabilisers.

5.5 Pitch Attitude to Collective Control

This test was to investigate the response to large sudden collective inputs maintaining all other control inputs. One can see from Figure 9 that upon a drastic increase in collective the aircraft actually enters a loop the loop cyclic state. Whilst it is unlikely that a sudden increase in collective will occur during the mission profile, this does highlight a massive instability of the aircraft. Caused by the reduction of the nose-down longitudinal flapping and hence reduction in nose down pitching moment due to collective pitch having a greater effect on the advancing side [10], a dramatic increase in collective will consequently result in a massive nose up moment.

**Figure 8:** Directional Step Impulse**Figure 9:** Pitch Attitude to Large Collective Inputs

5.6 Slung Load Test

Using an Xplane plugin [11], it was possible to model the flight dynamics effects of the aircraft with the rocket stage slung underneath. Whilst the slung load clearly affected the performance [11] of the helicopter, reducing its max speed to around 70 kts and greatly reducing its hovering capabilities, I found that it mostly increased the stability of the aircraft by further damping any oscillations. In fact if you observe Appendix A.3 and look at Figure 15 you can see that with the same magnitude of longitudinal input, the response is still oscillatory but not unstable. The magnitude of oscillations of the normal acceleration is also greatly reduced mostly likely as a result of the increased weight of the aircraft. This effect was observed with all other stability tests.

6 Oscillatory Requirements

Even though the aircraft may have stable oscillations when perturbed from equilibrium, there are still regulations and guidelines that recommend what frequency and damping ratio of the oscillation is safe. It is stipulated that the frequency w_n and damping ratio ζ meet the minimum requirements specified on Figure 10 upon a control pulse input in both roll and separately in yaw. Analysing the oscillations in Figure 8 one could find the natural frequency and damping ratio as 2.16 rad/s and 0.19 respectively. These could not be found for a unit roll input as this system was overly damped. Upon referring to Figure 10 we can see that these oscillations barely achieve Level 1 of the Cooper-Harper Handling Qualities Rating (See Appendix A.4 for the definition of these levels) and hence these oscillations are deemed safe and stable.

7 Qualitative Handling Response

Whilst quantitative analysis is very good at ensuring that the stability and performance of the aircraft meet the required benchmarks, it is still very important to observe qualitatively how it handles in the simulation. Important characteristics are manoeuvrability and ease of flight since it will be flown by a human pilot. Whilst these characteristics during capture have been noted by [11] I was interested in how it fared upon landing and takeoff in challenging conditions. One key observation is that with the lack of a tail rotor yaw control became very difficult to control in strong cross winds.

When a pilot comes into land with a strong cross wind, he uses the rudder pedals to apply a directional force to counter this crosswind and hence approach the helipad at side slip angle as shown in 11. A synchropter, however, has no rudder rotor and hence applies a directional force with differential collective. This is not as strong and hence is less able to counter strong crosswinds. Despite this, it must be noted that this aircraft is designed for the recovery of falling rocket stages, and a launch would not occur during difficult conditions therefore meaning the chances of this aircraft flying in challenging conditions is low. Additionally, a huge benefit of being launched from a ship is that the ship will be able to sail into the cross-winds essentially neglecting them (given that there is not too much lateral ship oscillation due to waves hitting the ship from the side). Whilst in Xplane the wake of the ship is not modelled, which is more critical for a frigate approach and less so for an approach onto a drill ship (which is what this aircraft will be launched from). This would be interesting for further analysis.

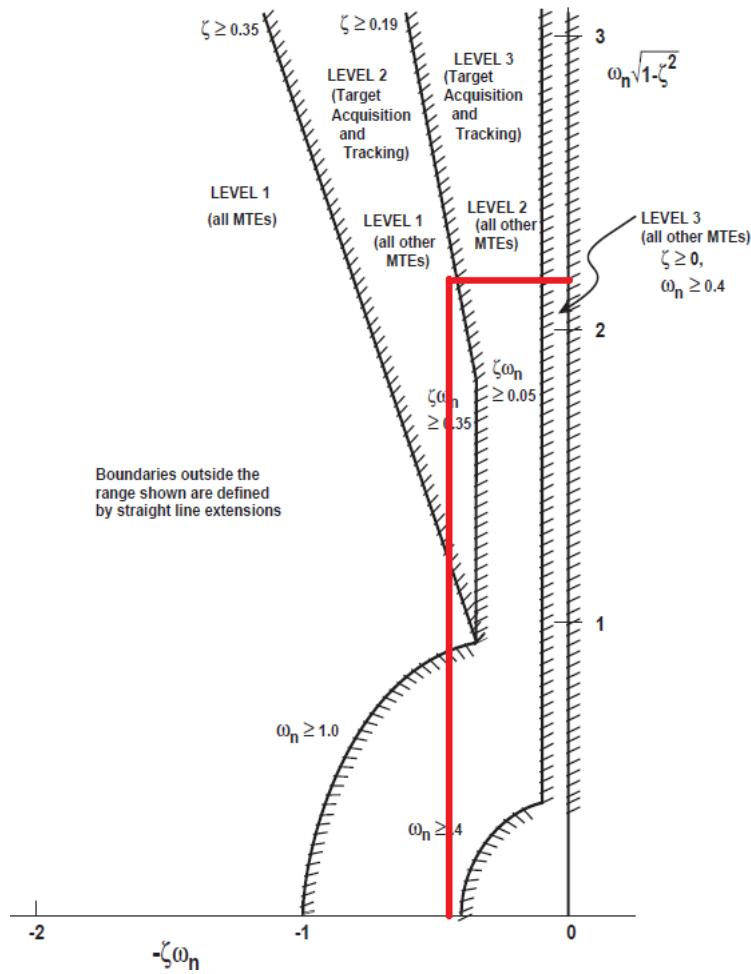


Figure 10: Lateral-directional Oscillatory Requirements



Figure 11: Final Approach of Frigate Landing in Stormy Conditions

8 Discussion

8.1 Limitations

One very apparent limitation to these tests was pilot inexperience. Whilst most tests required a step input given by the pilot through the cyclic joystick, it can be seen from various figures that the control input was merely a rough estimation of a step input. To reduce this limitation I practiced the test multiple times, however, with the lack of real time data I had to complete the test before viewing how successfully my attempt at achieving a step input was. This was further exacerbated by the lack of trim speed controls, which meant I had to estimate the original place I had held the joystick to maintain trim. There were also limitations with the flight simulation software. The most profound limitation was the fact that our model could only model twist whilst there was complex geometry, including sweep at the tip. This can greatly increase the aerodynamic characteristics [12] in particular the lifting performance. Whilst this has greater effect on performance tests [1] it would also change its oscillatory nature. The simulation team also found the Xplane 11 contained multiple bugs, and what would work on one computer would not work on another with the same set up. For example, the model the engine would not have any fuel flow on my computer but my other teammates, with the same settings and model, would not encounter this problem. This highlights the disadvantage of the black box style flight dynamics model of Xplane. Not being open source leads to some ignorance about how calculations are performed.

8.2 Revised Further Analysis and Design

Before this aircraft design is deemed to be appropriate to continue onto physical testing, I would suggest further revisions of the model and stability testing simulations. This is mainly directed at utilising the flight simulation software, in particular the plugins, in a better way to generate more accurate model and flight tests. By formulating trim speed controls and using a calculated computer (rather than user) input to conduct a control step input would eliminate pilot error. In terms of feedback to the design team to generate a better design, I would suggest that the aircraft had a greater ability of yaw control. Additionally, we have seen in the results that due to the high loads on the lightweight rotors due to low aspect ratio, I would also suggest a more sensitive throttle regulator be implemented rather than the default one Xplane uses. I would also suggest the implementation of some form of autopilot. I first tried to do the oscillatory tests in Xplane with Xplane artificial stability turned on, but was unable to achieve any oscillatory behaviour because the model was too stable. This highlights how having some form of control system in place can greatly increase the stability of the aircraft. Finally, and most importantly I would suggest that the aircraft has a stronger nose down pitching moment, as this is the biggest instability the aircraft faces.

9 Conclusion

- Conducting stability tests in the longitudinal, lateral and directional axes, the aircraft model implemented in Xplane was found to be stable.
- Sudden large increases in collective lead to a massive instability caused by an insufficient nose down pitching moment.
- Limitations included the inflexibility of the flight model as well as pilot inexperience.

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A Appendices

A.1 Longitudinal Engine Effects

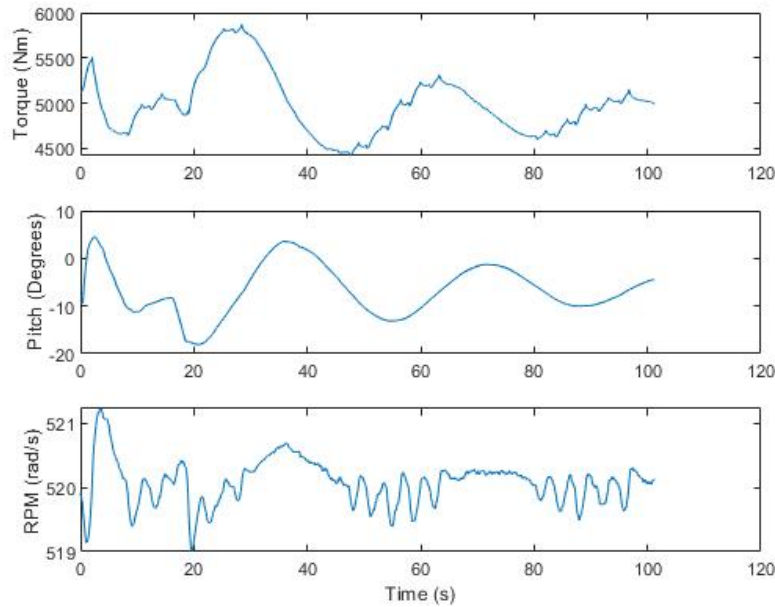


Figure 12: Engine Response Stable Longitudinal Stability

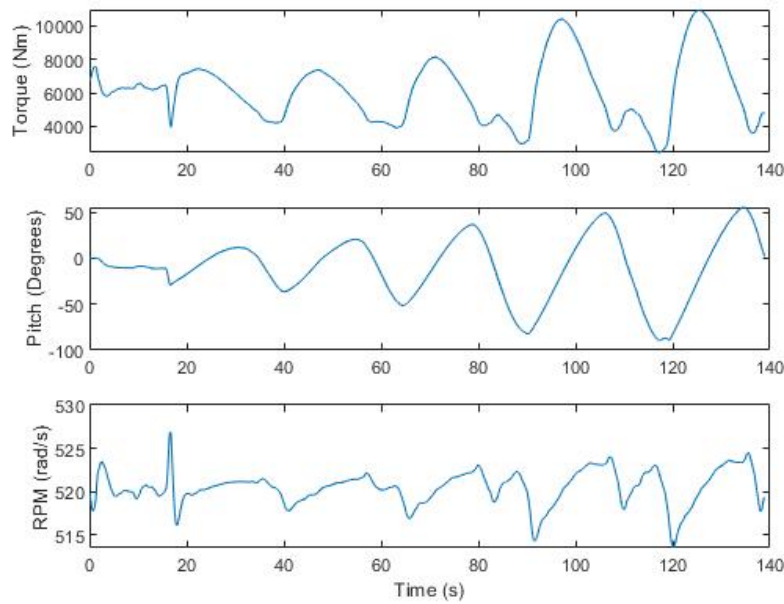


Figure 13: Engine Response Unstable Longitudinal Stability

A.2 Flow Fields

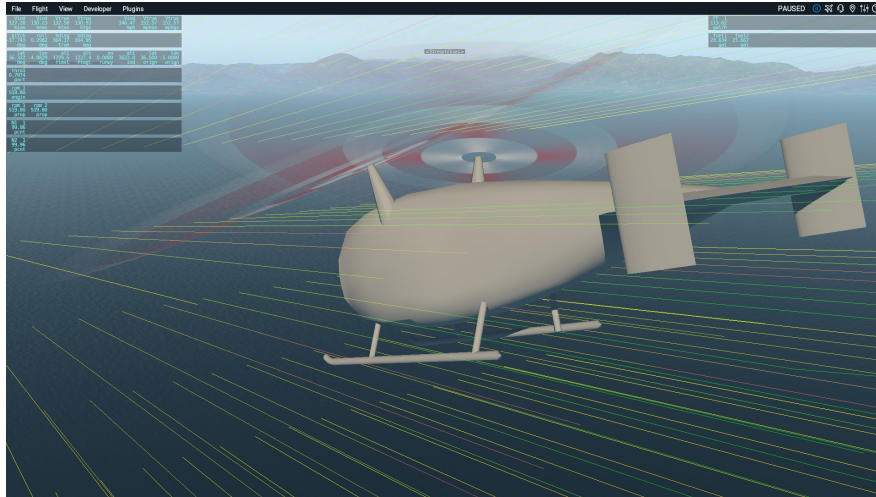


Figure 14: Streamlines Over Aircraft Travelling at 120 kts

A.3 Slung Load Stability

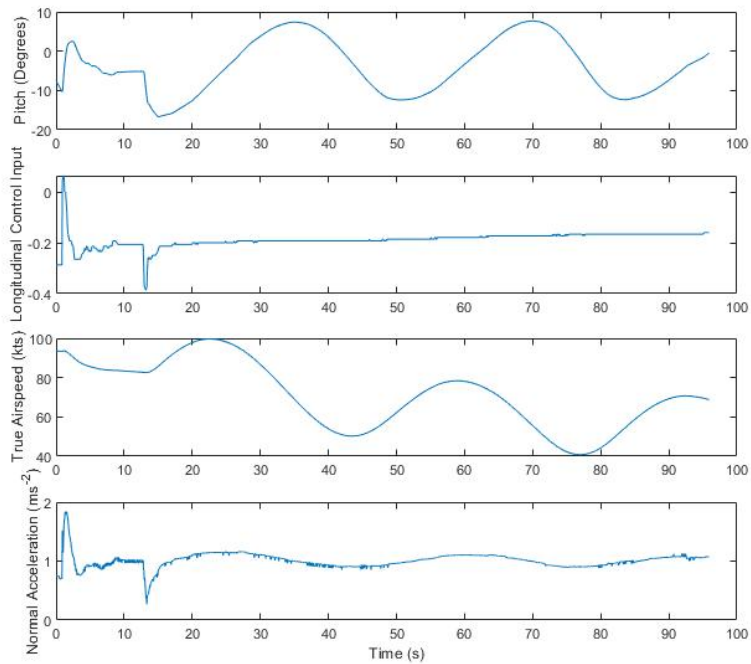


Figure 15: Slung Load Longitudinal Stability

A.4 Hooper-Harper Handling Qualities Rating

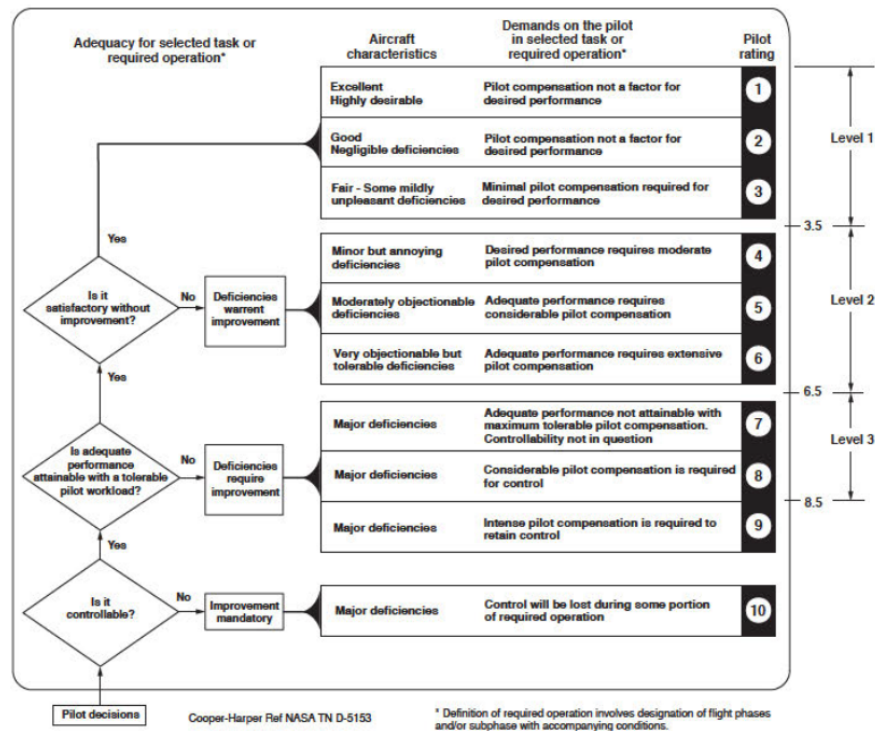


Figure 16: Hooper-Harper Handling Qualities Rating